J. Geomag. Geoelectr., **41**, 797–811, 1989

Paleomagnetic Study of the Tsushima Islands, Southern Margin of the Japan Sea

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(Received December 6, 1988; Revised May 9, 1989)

Paleomagnetic study of Oligocene and Miocene rocks from the Tsushima Islands, the southern margin of the Japan Sea, has revealed considerable deflection of the mean paleomagnetic direction compared with the Eurasian mean pole. Samples were collected from 48 sites of the late Oligocene to Early Miocene Taishu Group and Middle Miocene igneous intrusions with radiometric dates around 15 Ma. After demagnetization experiments, 14 sites from six stratigraphic units finally yielded resolvable stable magnetization components. An overall-mean direction of D: $-22.4°$, I: $44.7°$, and α_{95}: $11.7°$ was obtained. The mean declination was significantly deflected from that of the expected direction calculated from the 20 Ma Eurasian mean paleopole. The deflection, $-28.1° \pm 14.0°$, is explained by a counter-clockwise rotation of the Tsushima Islands relative to Eurasia after the igneous activity around 15 Ma. This result suggests that the Tsushima Islands were not included in the tectonic regime of the Southwest Japan block, which rotated clockwise during the formation of the Japan Sea at about 15 Ma.

1. Introduction

Since 1980, extensive paleomagnetic investigations have been carried out focusing on the rotational movements of Japanese island arcs (OTOFUJI and MATSUDA, 1983, 1984, 1987; HAYASHIDA and ITO, 1984; OTOFUJI *et al.*, 1985a, 1985b, 1985c; HAYASHIDA, 1986). These studies have revealed that a differential rotation took place between Southwest Japan and Northeast Japan in the Miocene time as a result of seaward rafting of continental fragments. The proposed tectonic event includes the breakup of the continental crust and subsequent back-arc spreading of the Japan Sea. The paleomagnetic directions show that Southwest Japan rotated clockwise through about 47° at 15 Ma relative to the stable Eurasia (OTOFUJI *et al.*, 1985c). The clockwise (CW) rotation of Southwest Japan necessitates a fan-shape opening of the western half of the Japan Sea. The pivot (pole) of the rotation has been roughly estimated at somewhere between the Korean

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Peninsula and Kyushu Island (OTOFUJI and MATSUDA, 1983).

The Tsushima Islands are located halfway between the Korean Peninsula and Kyushu Island at the southern margin of the Japan Sea (Fig. 1). Paleomagnetic directions of the Cenozoic rocks on the islands enable us to characterize the tectonic movements associated with the formation of the Japan Sea. Before now, paleomagnetic directions were available only from the Middle Miocene granitic rocks on the islands. The characteristic magnetization directions of the granite show counterclockwise (CCW) deflection from the geocentric axial dipole field (ITO *et al.*, 1980). The CCW-deflected direction may suggest that the Tsushima Islands were not included in the tectonic regime of the Southwest Japan block.

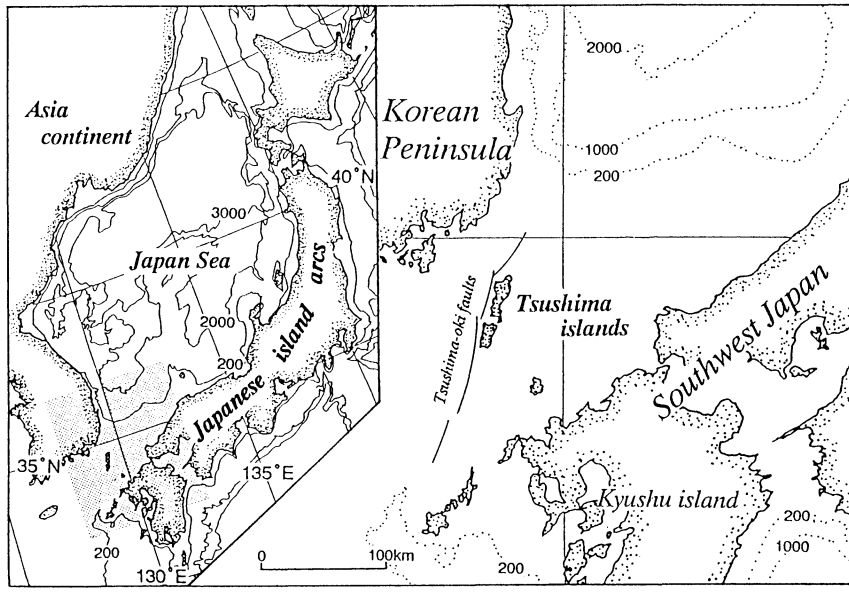


Fig. 1. Index map of the Tsushima Islands (modified from GEOLOGICAL SURVEY OF JAPAN (1982)).

2. Geologic Setting and Sampling

The Tsushima Islands are comprised of two main islands (Kamishima to the north and Shimojima to the south) and a number of islets. Thick marine sediments of late Oligocene to early Miocene (the Taishu Group) are exposed throughout the islands (Fig. 2(a)). Various plutonic extrusive bodies are distributed as dikes, sills and volcaniclastics on the islands. The Taishu Group is stratigraphically subdivided into three formations, the Lower, Middle, and Upper Formations (e.g., SHIMADA, 1977; Fig. 3). Intercalations of thin volcaniclastic beds occur in the lowermost part of the Middle Formation. The Taishu Group is characterized by intensive folding of

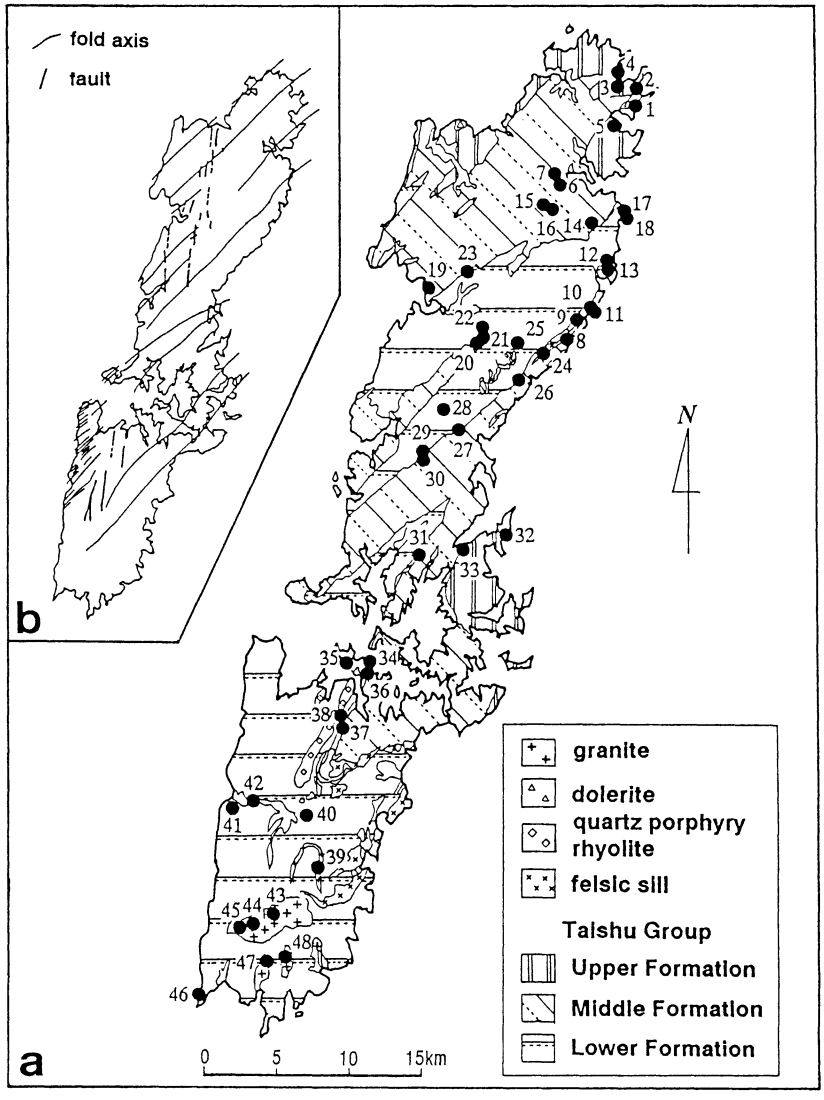


Fig. 2. (a) Geologic map of the Tsushima Islands (simplified from SHIMADA (1977) and KOGA (1982)). Solid circles with numerals indicate locations of paleomagnetic sampling sites. (b) Map showing simplified geologic structure (modified from TOMITA *et al.* (1975)).

its strata (Fig. 2(b)); the fold axes generally trend NE-SW, plunging gently at about 10° to the northeast. There are many faults truncating the fold system, we cannot find any large scale faults running across the islands. The distribution of the Taishu Group and the fold system is continuous throughout Kamishima and Shimojima. The Tsushima Islands can be regarded as a tectonically coherent block.

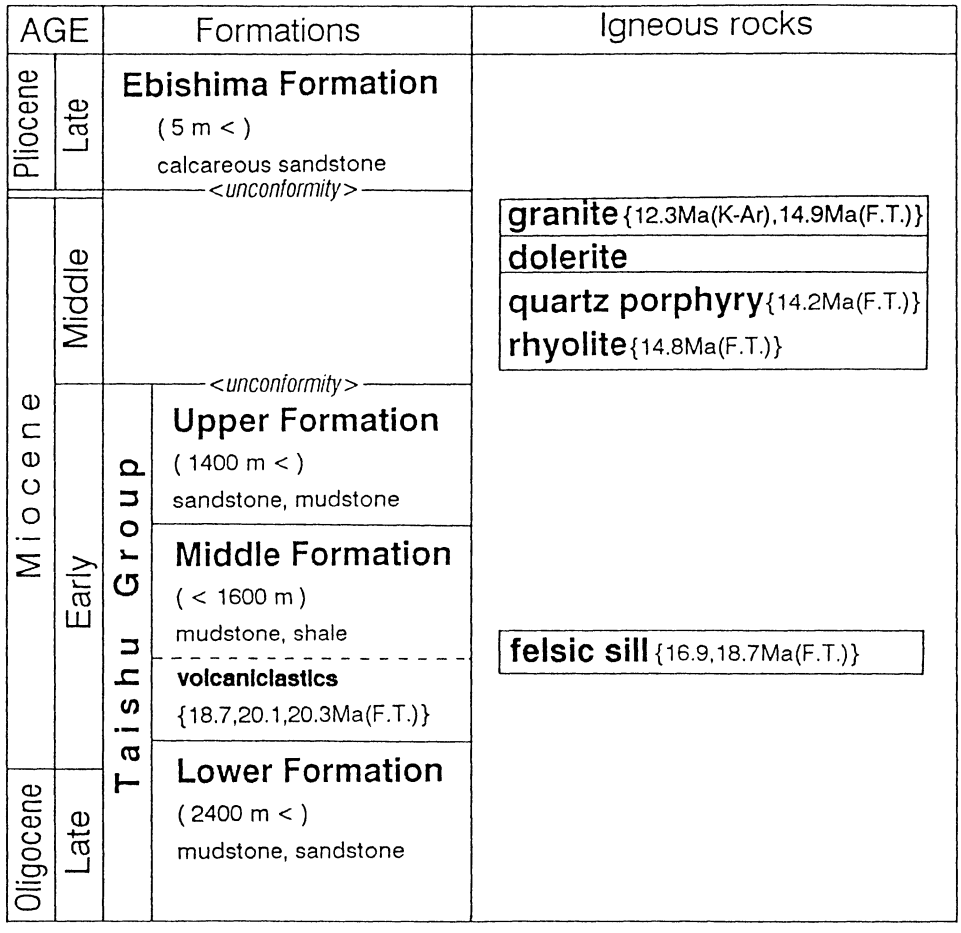


Fig. 3. Stratigraphic relations of sedimentary and igneous rocks in the Tsushima Islands. Radiometric ages reported so far are shown.

Felsic igneous rocks occur as plutons, sills, dikes and lava flows. Mafic rocks (dolerite) form dikes. The felsic sills are dominantly intruded into the lower horizon of the Middle Formation, close to the volcaniclastic layers and are folded concordantly with the Taishu Group. The quartz porphyry, rhyolite, dolerite and granite intersect the fold system of the Taishu Group. Intrusion of the granite occurred at the latest stage of igneous activity on the islands (MITI, 1972, 1973, 1974; SHIMADA, 1977).

The stratigraphic relationship among the rock units on the Tsushima Islands is illustrated in Fig. 3. The geologic age of the Taishu Group has been estimated to range from late Oligocene to early Miocene, based on plant and molluscan fossils

(e.g., SHIMADA, 1977). Radiometric ages of the volcaniclastic rocks were determined by the fission-track method (18.7±1.2, 20.1±1.2, and 20.3±1.5 Ma: TAKAHASHI and HAYASHI, 1985, 1987). These dates support the paleontological age estimation of the Taishu Group. The folded felsic sills yielded fission-track ages of 16.9±1.0 and 18.7±1.0 Ma (TAKAHASHI and HAYASHI, 1985, 1987). The radiometric ages of the post-folding, igneous rocks have been reported as a single K–Ar date of 12.3 Ma by KAWANO and UEDA (1966) (recalculated by the authors due to the revision of radiometric constants (STEIGER and JAĞER, 1977)) and three fission-track ages of 14.2±0.7, 14.8±0.8, and 14.9±0.8 Ma by TAKAHASHI and HAYASHI (1985, 1987).

The samples for the paleomagnetic study were collected at 48 sites distributed over the whole area of the islands: 35 sites from the Taishu Group and 13 sites from the igneous rocks (Fig. 2(a)). At each site, about ten block samples were collected and oriented using a magnetic compass. For the sedimentary rocks, we measured strike and dip angles of the strata. In the case of felsic sills, we measured the attitude of bedding of the country rocks adjacent to the sills.

3. Paleomagnetic Analysis

3.1 Demagnetization experiments

Two or more cylindrical specimens 25 mm in diameter and 22 mm in height were prepared from each hand sample for the magnetic measurement. Remanent magnetism was measured on a cryogenic magnetometer (ScT C-112) and/or on a spinner magnetometer (Schonstedt SSM-1A). The stability of the natural remanent magnetization (NRM) was examined through progressive demagnetization experiments by the thermal and alternating field (AF) methods. Two or more pilot specimens from each site were subjected to progressive demagnetizations. Individual magnetization components for each pilot specimen were picked on the basis of straight line trends of vector-end points toward the origin of the vector demagnetization diagram (ZIJDERVELD, 1967).

The NRM of pilot specimens from most of the sedimentary sites of the Taishu Group showed an irregular behavior during the progressive demagnetizations, especially at treatment temperatures above 300°C and/or peak fields above 10 mT (Figs. 4(a) and (b)). Only two sites out of 35 of the Taishu Group, namely site 34 (volcaniclastics) and site 42 (shale of the Lower Formation), yielded stable components which were recognized as straight line trends toward the origin on the diagram. The stable components of the two sites were resolved only by thermal demagnetization at temperatures above 500°C (Figs. 4(c), (d), (e), and (f)).

All pilot specimens from the igneous rocks (13 sites) yielded stable components by progressive thermal and/or AF demagnetization experiments. Two different types of demagnetization character were observed during the progressive demagnetization experiments. The remanent magnetism of site 24 (felsic sill), 32 (felsic sill), 38 (quartz porphyry), 43 (granite), 44 (granite), and 47 (granite) essentially comprised a single, stable component with a soft component which was easily removed by low-demagnetization treatments below about 200°C and/or 10 mT (Figs. 4(g) and (h)).

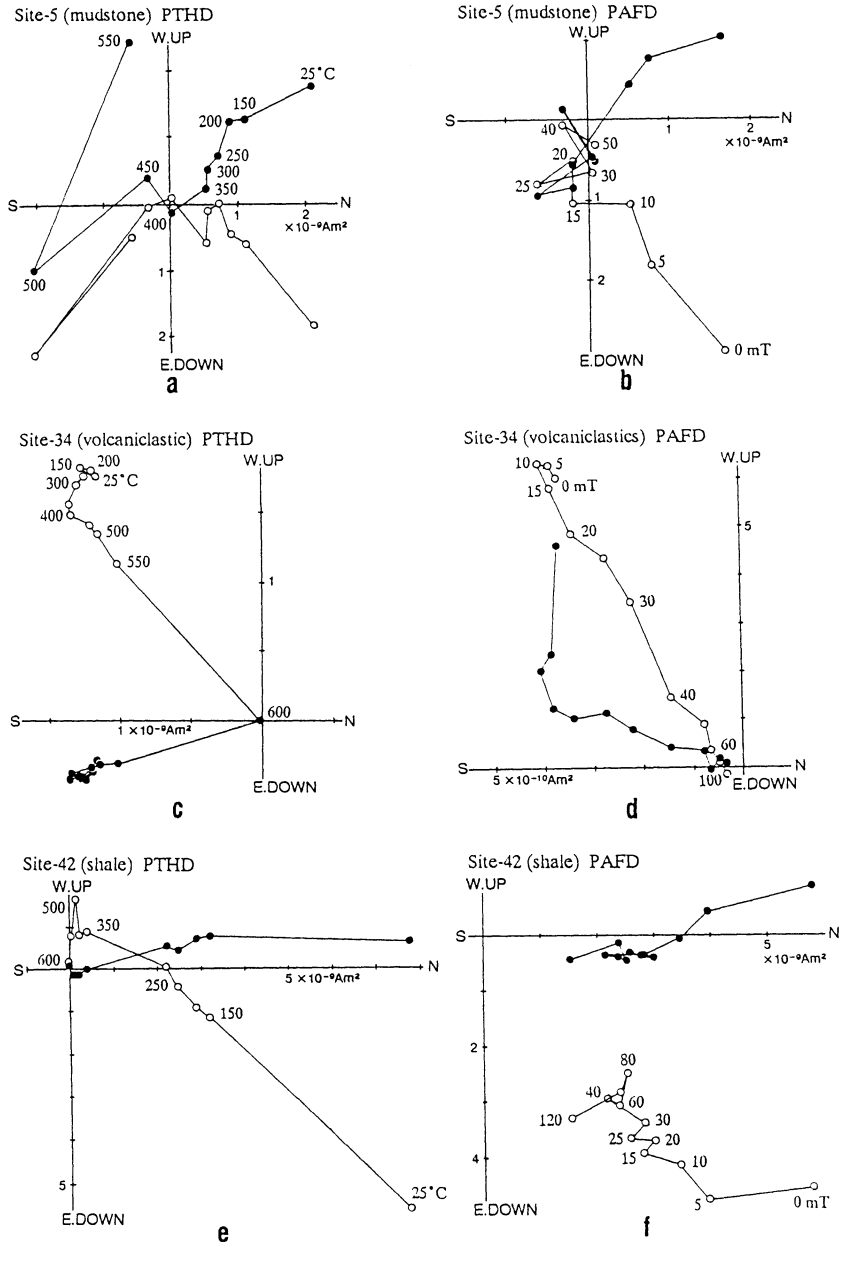

Site-5 (mudstone) PTHD
W.UP
550
25°C
150
200
250
300
350
450
400
500
S
N
1
2
×10⁻⁹Am²
E.DOWN
a
Site-5 (mudstone) PAFD
W.UP
40
50
20
30
25
15
10
5
0 mT
E.DOWN
b
Site-34 (volcaniclastic) PTHD
W.UP
150
200
300
25°C
400
500
550
600
1 ×10⁻⁹Am²
E.DOWN
c
Site-34 (volcaniclastics) PAFD
W.UP
10
5
15
0 mT
20
30
40
60
100
5 ×10⁻¹⁰Am²
E.DOWN
d
Site-42 (shale) PTHD
W.UP
500
350
600
250
150
5 ×10⁻⁹Am²
25°C
E.DOWN
e
Site-42 (shale) PAFD
W.UP
5
×10⁻⁹Am²
80
40
60
120
30
25
20
15
10
5
0 mT
E.DOWN
f

Fig. 4.

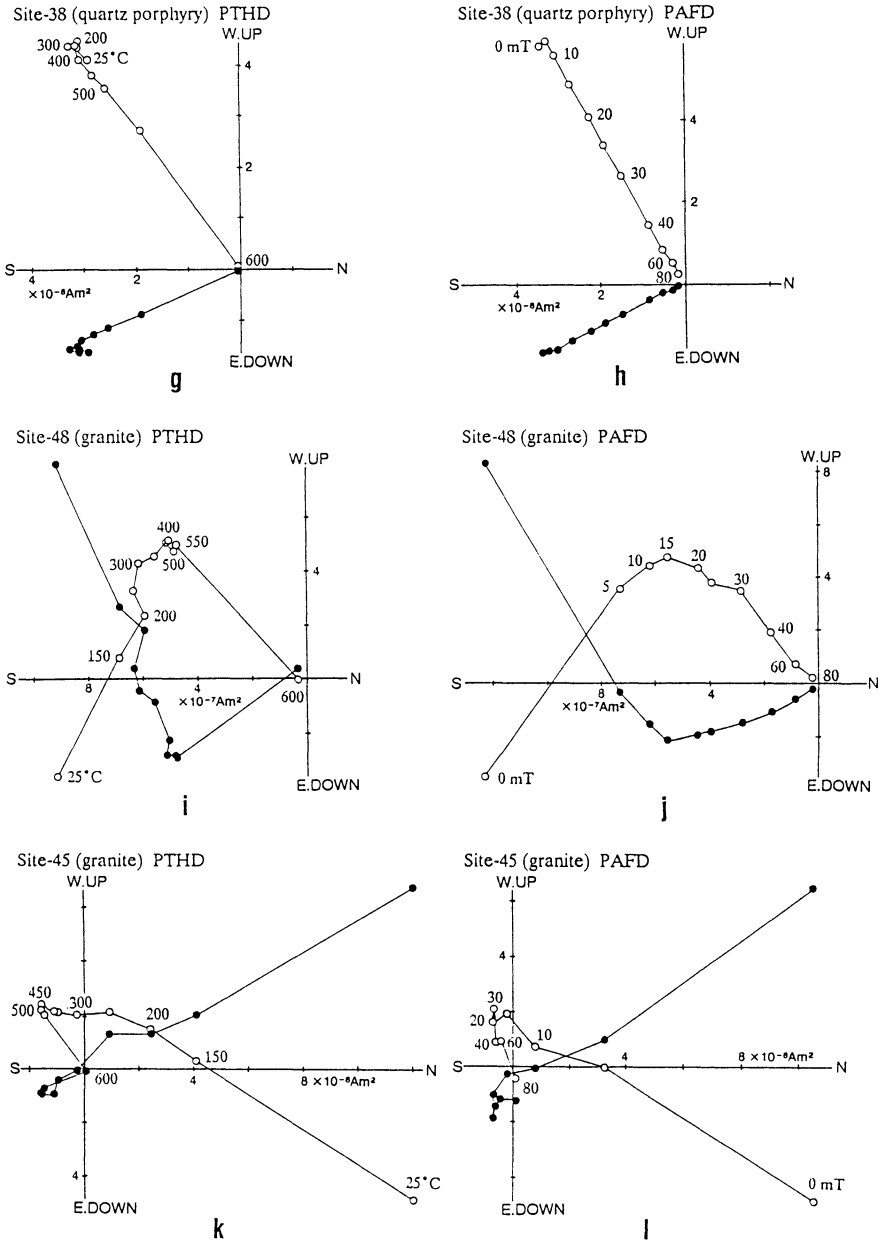


Fig. 4. Typical examples of progressive demagnetization experiments shown by vector-demagnetization diagram. Solid and open symbols are projections on the horizontal and N-S vertical planes, respectively. Coordinate unit is bulk remanent intensity. PTHD: progressive thermal demagnetization, PAFD: progressive alternating field demagnetization. (a), (b): mudstone of the Upper Formation of Taishu Group, (c), (d): volcaniclastic rock in the lowermost part of the Middle Formation of Taishu Group, (e), (f): shale of the Lower Formation of Taishu Group, (g), (h): quartz porphyry, (i), (j), (k), (l): granite.

The rest of the sites, site 1 (dolerite), 10 (felsic sill), 25 (dolerite), 28 (dolerite), 39 (granite), 45 (granite), and 48 (granite), had stable components with high blocking temperatures and high coercivity. In sites 10 and 48, the stable components were recognized as a straight line trend by both thermal and AF methods, in the higher-temperature range above 400 or 500°C and in the higher peak-field above 20 or 30 mT (Figs. 4(i) and (j)). Except for sites 10 and 48, the thermal treatment was more effective than the AF treatment in isolating the stable components (Figs. 4(k) and (l)).

All samples were demagnetized at several levels in the demagnetization interval which yielded the stable component in the respective pilot sample. Thermal treatments were mainly used. The samples of five sites (sites 24, 43, 44, 47, and 48) which revealed the stable components by the AF treatment were not thermally demagnetized. The site-mean directions of the stable sites were calculated for each level in the optimal range of demagnetization. The demagnetization step within this range which gave the mean direction with the minimum dispersion was selected as the characteristic mean direction of the site. The site-mean directions were obtained at 15 sites. The radius of 95% confidence limits (α_{95}) of these site-mean directions were all smaller than 30° (Fig. 5 and Table 1).

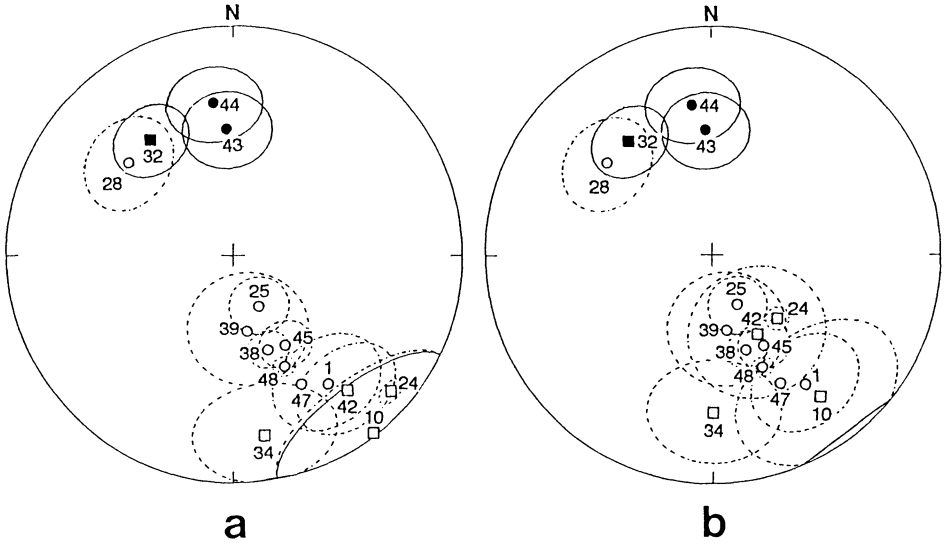


Fig. 5. Equal-area projection of paleomagnetic site-mean directions; (a): in situ, (b): after tectonic corrections. Circles are the directions from the post-folding rocks and squares from the pre-folding rocks. Ovals around the directions indicate 95% confidence limit. Numerals denote site number. Solid and open symbols are on the lower and upper hemispheres, respectively.

Table 1. In situ (D, I) and tectonic-corrected (D^*, I^*) site mean directions.

Site	Level	D	I	D^*	I^*	α_{95}	k	N
Igneous rocks								
Granite [Gr]								
39	500°C	169.4	−62.5			21.0	7.9	8
43	15 mT	−3.0	44.6			14.9	12.9	9
44	15 mT	−7.4	34.1			15.4	9.7	11
45	500°C	150.2	−52.4			9.1	38.2	8
47	30 mT	151.8	−36.3			1.9	854.0	8
48	20 mT	156.0	−45.2			3.8	184.3	9
mean		−17.2	46.3			11.1	37.6	(6)
Dolerite [Dl]								
1	500°C	144.2	−30.8			18.0	12.1	7
25	550°C	153.4	−69.7			10.5	22.1	10
28	550°C	−49.5	−38.3			16.3	8.8	11
mean (1, 25)		−33.2	50.3			—	—	(2)
Quartz porphyry [Qp]								
38	400°C	160.0	−53.4			7.1	47.7	10
Felsic sill [Fs]								
10	550°C	142.5	−1.1	143.0	−23.1	26.7	9.1	5
24	30 mT	131.6	−10.2	134.3	−57.1	4.1	155.5	9
32	400°C	−36.5	38.2			13.3	21.7	7
mean		−41.1	16.5			31.9	16.0	(3)
mean*				−38.9	39.5	27.0	21.8	(3)
Taishu Group								
volcaniclastics [vo]								
34	500°C	170.3	−20.4	179.1	−30.8	21.2	9.1	7
Lower Formation shale [sh(L)]								
42	500°C	140.5	−23.9	150.5	−57.2	16.6	17.3	6

α_{95}: radius of 95% confidence circle in degrees. k=precision parameter. N=number of specimens. mean, mean*: mean direction of each rock unit calculated from site mean directions before and after tectonic corrections, respectively.

3.2 Results

We obtained site-mean directions from five sites of pre-folding rocks (sedimentary rocks of the Taishu Group and the felsic sills) and ten sites of post-folding, younger igneous rocks (Table 1). The directions of the folded rocks were subjected to tilt correction. Unplunging and unfolding of the remanent vectors were calculated based on structural data of the Taishu Group. We estimated the plunge angles of each fold axis from detailed geologic maps (MITI, 1972, 1973, 1974) and our field observation. As axial planes of the folding system were approximately vertical at each site where we obtained the stable site-mean direction, a simple tilt correction after the unplunging could produce a proper tectonic correction. On the other hand, we could not make any structural correction for the post-folding, younger igneous rocks. The igneous activity of those rocks has been considered to

postdate the folding event of the Taishu Group (e.g., MITI, 1974); we thus simply assumed that the younger igneous rocks had not suffered significant tilting.

In performing the tilt correction for the directions from the pre-folding rocks, four site mean directions (sites 10, 24, 34, and 42) change in inclination from extremely shallow to moderate (Fig. 5). The direction of site 32 does not change because the attitude of the bedding at site 32 is almost horizontal. Among the five site-mean directions of the pre-folding rocks, three directions from the felsic sills (sites 10, 24, and 32) show rather a better antiparallel relationship after unfolding (α_{95} decreases from 31.9° to 27.0°). The directional scatter of five unfolded directions is almost similar to that of the in-situ directions (α_{95} of a mean direction calculated from in-situ directions is 19.0° and that from unfolded directions is 19.7°). Unsuccessful fold tests may be attributed to the small number of the sites of stable remanence. After tilt correction, each direction of the pre-folding rocks shows more moderate inclination, which is close to that of the expected geomagnetic field direction in Oligocene and Miocene times at the Tsushima Islands. We thus adopt the unfolded directions of the pre-folding rocks as their primary magnetizations.

All site-mean directions from igneous rocks, except that of site 28, make two tight antipodal clusters in the NW quadrant of the lower hemisphere and the SE quadrant of the upper hemisphere of the equal-area projection (Fig. 5). These directions are concordant with the unfolded directions of the folded rocks.

After tilt correction, the antiparallel relationship is observed more clearly throughout the site-mean directions obtained from various stratigraphic units (Fig. 5(b)). The antiparallel clusters can be assumed to be a record of the normal and reversed geomagnetic fields. The direction of site 28 is anomalous and may be a record of the transitional field during a geomagnetic reversal. Site 28 is discarded from further consideration.

Stable, tilt-corrected remanent directions are determined on 14 sites from six stratigraphic units of the island. An overall-mean direction is thus obtained by combining six formation-means into one: the mean declination and inclination are −22.4° and 47.1°, respectively. The radius of 95% confidence limit is 11.7° and the precision parameter (k) is 33.9 (Fig. 6, Table 2). This direction can be regarded as the mean paleomagnetic direction for the late Oligocene to Middle Miocene in the Tsushima Islands.

4. Discussion and Interpretation

ITO *et al.* (1980) previously reported the CCW deflection of remanent directions of granitic rocks from the southern part of Shimojima. We also collected samples at six sites from the same granite bodies, and reconfirmed CCW deflection of mixed polarity. ITO *et al.* (1980) stressed the possibility of the recording of a transitional field during a geomagnetic reversal based on the presence of both normal and reversed polarities. The present study, however, revealed that the CCW deflection is observed throughout the Oligocene to Miocene rocks on the islands. It follows that instantaneous recording of the transitional field cannot be accepted to explain the

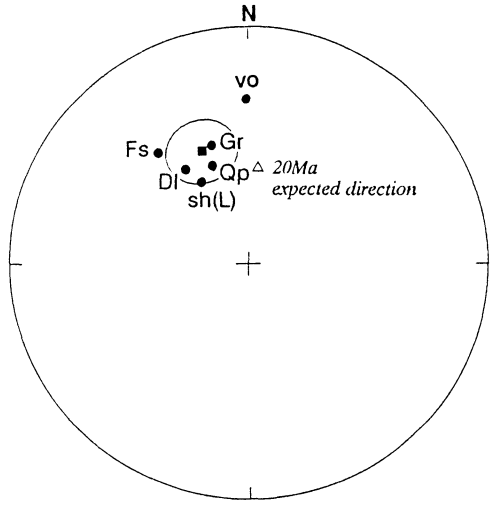


Fig. 6. Equal-area projection of formation-mean directions after tectonic corrections (see Table 2). Square indicates an overall-mean direction. Oval indicates 95% confidence limit. Triangle indicates expected magnetization direction for the Tsushima Islands calculated from a 20 Ma Eurasia mean paleopole (IRVING and IRVING, 1982).

Table 2. Paleomagnetic formation-mean directions and an overall-mean direction.

Rock unit	N^*	D	I	D^*	I^*	α_{95}	k
Gr	6	−17.2	46.3			11.1	37.6
Dl	2	−33.2	50.3			—	—
Qp	1	−20.0	53.4			—	—
Fs	3	−41.1	16.5			31.9	16.0
				−38.9	39.5	27.0	21.8
vo	1	−19.7	20.4	−0.9	30.8	21.2	9.1
sh(L)	1	−39.5	23.9	−29.5	57.2	16.6	17.3
mean	6	D=−22.4		I=47.1		11.7	33.9

Gr: granite, Dl:dolerite, Qp: quartz porphyry, Fs: felsic sill, vo: volcaniclastic rock in the lowermost part of the Middle Formation of Taishu Group, sh(L): shale of the Lower Formation of Taishu Group (see Table 1). N^*: number of sites.

anomalous paleomagnetic directions.

The CCW deflection of the paleomagnetic direction can be interpreted in terms of the tectonic displacement of the whole Tsushima Islands. We took the 20 Ma Eurasian mean paleopole (IRVING and IRVING, 1982) as a reference and then calculated the expected direction at the Tsushima Islands (34.3°N, 129.3°E). The declination and inclination of the expected dipole field is 5.7° and 55.6°, respectively. Anomalies of declination and inclination with 95% confidence limit were

based on the definitions of BECK (1980) and DEMAREST (1983). The anomalies of declination and inclination are $-28.1° \pm 14.0°$ and $8.5° \pm 9.6°$, respectively. While the inclination anomaly is not significant, the declination anomaly is significant. The negative anomaly of the declination can be explained as a counterclockwise (CCW) rotation of the Tsushima Islands through about 28° around a vertical axis relative to Eurasia.

The CCW rotation of the Tsushima Islands postdated the intrusion of granite, which was the last stage of the igneous activity on the Tsushima Islands. The radiometric ages of the granite are slightly discordant: a 12.3 Ma K–Ar date with 5% error (KAWANO and UEDA, 1964, 1966), and a 14.9 ± 0.9 Ma fission track date (TAKAHASHI and HAYASHI, 1985). Since the K–Ar dating was carried out over twenty years ago and the fission track dating is not calibrated to the age standard (HURFORD and GREEN, 1982; MILLER *et al.*, 1985), the apparent discrepancy between two dates is not statistically significant. The emplacement of the granite body may be about 15 Ma or slightly younger. Concentration of the fission track dates of the other igneous rocks to 15 Ma may support a single magmatic activity at 15 Ma (TAKAHASHI and HAYASHI, 1985, 1987; Fig. 2). We can safely say that the CCW rotation of the Tsushima Islands had occurred at sometime after 15 Ma.

Our paleomagnetic result suggests that the Tsushima Islands did not belong to the CW-rotated Southwest Japan, and a different tectonic role for the islands in the formation process of the Japan Sea is suggested. Around the southern margin of the Japan Sea, from on the southeastern coast of the Korean Peninsula and in the sea area off the northwestern coast of Kyushu Island, several faults with mainly NNE-SSW trends have been found by geological surveys on land and marine seismic investigations (e.g., INOUE, 1982; CHOI and PARK, 1985). Among these remarkable faults are the Tsushima-oki faults (TOMITA *et al.*, 1975), which is a large-scale fault zone lying off the western coast of the Tsushima Islands (MURAUCHI, 1972; TOMITA *et al.*, 1975; KATSURA and NAGANO, 1976). The faults had an effect on the deformation of the strata on the islands (TOMITA *et al.*, 1975; SHIMADA, 1977; INOUE, 1982); the sedimentary rocks on the western coast of the islands are more intensively deformed than those of the eastern part, and the fold axes are dragged along the western coast line and change in trend from NE-SW to NNE-SSW. These geological features imply a sinistral motion of the Tsushima-oki faults (SHIMADA, 1977; INOUE, 1982). In addition to the Tsushima-oki faults, KATSURA and NAGANO (1976) pointed out the existence of a NE-SW fault off the eastern coast of the Tsushima Islands from the results of seismic surveys, and regarded the Tsushima Islands as a faulted block. The CCW rotation of the Tsushima Islands may have taken place in response to the movement of the fault system at the southern margin of the Japan Sea, especially to the sinistral motion of the Tsushima-oki faults. The CCW rotation may result from a plastic drag near the sinistral faults or the rotation of a faulted block in a left-lateral shear zone (MACDONALD, 1980). The time of the movement of the Tsushima-oki faults seems to be at syn- or post-folding (SHIMADA, 1977; INOUE, 1982). The time of the folding event is bracketed by the radiometric dates on the folded felsic sills and non-folded igneous rocks; it is between 19 Ma and

15 Ma. Hence, the time of active fault movement may have been at or after 15 Ma, which is concordant with the CCW rotation of the Tsushima Islands suggested from the present paleomagnetic study.

The climax of the opening event of the Japan Sea was dated at 15 Ma on the basis of the well-documented paleomagnetic data in conjunction with the geochronological evidence. The time of the CCW rotation of the Tsushima Islands is close to 15 Ma if we take the older limit of the possible age range discussed above. The fan-shape opening of the southern part of the Japan Sea should have caused both the CW rotation of Southwest Japan and the CCW rotation of the Tsushima Islands, possibly in response to the fault movements.

5. Conclusion

1) Stable, tilt-corrected directions were determined on 14 of 48 sites from six stratigraphic units. A mean paleomagnetic direction from the six formation-mean directions was $D=-22.4°$, $I=47.1°$, and $\alpha_{95}=11.7°$, which was regarded as the mean paleomagnetic direction for the late Oligocene to Middle Miocene in the Tsushima Islands.

2) The paleomagnetic direction of the islands shows a significant declination anomaly of $-28.1° \pm 14.0°$ when compared to the 20 Ma reference pole for Eurasia (IRVING and IRVING, 1982), which suggests a counterclockwise (CCW) rotation about a vertical axis of about 28° relative to Eurasia. The CCW rotation of the islands took place after the igneous activity at about 15 Ma.

3) The Tsushima Islands did not belong to the Southwest Japan block during the formation process of the Japan Sea.

We thank A. Hayashida for his help in the field and his critical reviewing of the manuscript. Microscopic observation of thin sections and discussion by Y. Tatsumi are appreciated. Discussions with T. Tagami and S. Nishimura were very helpful. J. Channell greatly enhanced the manuscript and anonymous reviewers made many helpful comments. This work was supported in part by a Grant-in-Aid (No. 60540491) from the Japanese Ministry of Education, Science and Culture.

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